# SHANG WU

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## RESEARCH INTERESTS

I am broadly interested in human–AI interaction, AI evaluation and benchmarking, and uncertainty quantification. I develop statistical and cognitively inspired methods to evaluate and improve AI systems. My work spans Bayesian and sequential evaluation of black-box LLMs for reliable, cost-efficient benchmarking; uncertainty calibration and how LLMs express certainty in natural language; and optimal task allocation across human and AI agents under real-world constraints. I design user studies on how AI assistance shapes human reasoning, learning, and reliance on deep learning systems.

## EDUCATION

**Ph.D. in Computer Science** **09/2023 – Expected 06/2028**
University of California, Irvine
Advisor: Padhraic Smyth

**M.S. in Computer Science** **09/2023 - 03/2025**
University of California, Irvine | GPA 3.96/4.00

**B.S. in Computer Science and Psychology with Honors** **09/2020 - 03/2023**
University of California, Davis | GPA 4.00/4.00

## PUBLICATIONS

- Belém, C., **Wu, S.**, Yao, H., Steyvers, M., Singh, S., Smyth, P., From 'May' to 'Is': Certainty Distortion in Language Model Rewriting. *EMNLP 2026.*
- **Wu, S.**, Belém, C., Fu, S., Steyvers, M., Smyth, P., How AI Assistance Affects Human Skill Development: A Study of Learning with Logic Puzzles. *HCOMP 2026.*
- **Wu, S.**, Yao, H., Belém, C., Fu, S., Steyvers, M., Smyth, P., The Impact of AI Usage and Informativeness on Skill Development in Logical Reasoning. *HHAI 2026.*
- Longjohn, R., **Wu, S.**, Kher, S., Belém, C., Smyth, P., Bayesian Evaluation of Black-box LLM Behavior. *NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle.*

## WORKING PAPERS

- **Wu, S.**, Kher, S., Smyth, P., Learning to Assign Prediction Tasks to Agents with Capacity Constraints. *Under Review.*
- Kher, S.*, **Wu, S.***, Longjohn, R., Belém, C., Smyth, P., Sequential Bayesian Evaluation of Large Language Model Behavior. *Under Review.* (* Equal contribution)
- **Wu, S.**, Yao, H., AI Evaluation May Bias Perceptions: The Importance of Context in Interpreting Academic Writing. *Under Review.*

## RESEARCH EXPERIENCE

**University of California, Irvine | Graduate Student Researcher** **09/2023 – Present**
Computer Science Department; Advisor: Padhraic Smyth

- **Human–AI Interaction & Collaboration:** Led end-to-end study design and empirical analysis of user studies on how AI assistance shapes human learning, reliance, and decision-making; developed machine learning methods to optimally allocate tasks between humans and AI under real-world capacity constraints.
- **LLM Evaluation & Benchmarking:** Developed Bayesian and sequential evaluation frameworks for reliable, sample-efficient benchmarking of black-box deep learning models, including LLMs, with calibrated uncertainty; conducted large-scale empirical studies of how LLMs express uncertainty.

**University of California, Davis | Research Assistant**
Psychology Department 06/2022 - 03/2023

- Designed and implemented agent-based simulation models in Java to study how publication bias and research incentives inflate effect sizes, bridging computational modeling and behavioral science.

Animal Science Department 02/2022 - 05/2022

- Conducted data analysis and visualization in Python and R for genomics research publications.

## FELLOWSHIPS & AWARDS

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| HPI Research Center Fellowship | 01/2025 |
| UCI ICS Dean's Fellowship | 06/2024 |
| Thomas E. Bruzzone Scholarship | 12/2022 |
| Charles R. Bishop Award | 09/2022 |
| 2021 Michael's Scholarship | 01/2022 |

## SKILLS

- **AI & ML Methods:** LLM evaluation and benchmarking, uncertainty quantification and calibration, Bayesian inference, probabilistic modeling, sequential evaluation, human-in-the-loop task allocation, prompt engineering, and deep learning.
- **Empirical Research:** Experimental design, A/B testing, causal inference, large-scale model evaluation, user studies and crowdsourced data collection, qualitative and quantitative error analysis, statistical modeling, and behavioral data analysis.
- **ML Frameworks:** PyTorch, Hugging Face Transformers, scikit-learn, Stan, and LLM APIs.
- **Programming:** Python, R, NumPy, pandas, matplotlib, C++, Java, Git, and LaTeX.
- **Languages:** English and Mandarin.

## TALKS & PRESENTATIONS

**Wu, S.**, Understanding Human–AI Dynamics in Interactive Decision-Making Tasks, *Collective Intelligence 2025 Demos and Posters.*

## TEACHING

**Student Assistant - UC Davis Computer Science Department** **04/2022 - 06/2022**
Course: Software Development & Object-Oriented Programming in C++.

**Peer Tutor – UC Davis Academic Assistance and Tutoring Centers** **10/2021 - 03/2023**
Programming, mathematics, and statistics.